Modeling Human Activity States Using Hidden Markov Models

Formative 2 · HMM Human Activity Recognition

Emmy Nshimiye | Blessing Hirwa

GitHub Repository: https://github.com/nshimiyeemmy/Formative-2_Hidden_Markov_Model/

1. Background and Motivation

Human Activity Recognition (HAR) is fundamental to modern pervasive computing, enabling applications in healthcare monitoring, sports science, and smart environments. This project tackles a **wearable-based wellness monitoring** use case: automatically detecting four activities—standing, walking, jumping, and remaining still—from smartphone IMU signals alone. The key challenge is that the true activity is *hidden*; only noisy accelerometer and gyroscope readings are observable. Hidden Markov Models (HMMs) are ideally suited to this setting because they explicitly model the probabilistic relationship between latent activity states and sensor observations, while also capturing the temporal dynamics of activity transitions.

2. Data Collection and Preprocessing

2.1 Collection Protocol and Sampling Rate Harmonization

Data were collected using **Sensor Logger** on two Android smartphones with different native sampling rates (~100 Hz and ~200 Hz). nshimiyeemmy recorded **jumping and still** sessions; hirwaalu recorded **standing and walking** sessions. Each activity was recorded across multiple 5–10 s sessions, producing more than 50 labelled files. Both accelerometer and gyroscope streams were resampled to a unified **50 Hz** target using linear interpolation. This rate satisfies the Nyquist criterion for human motion (informative content < 20 Hz) and ensures device-agnostic consistency.

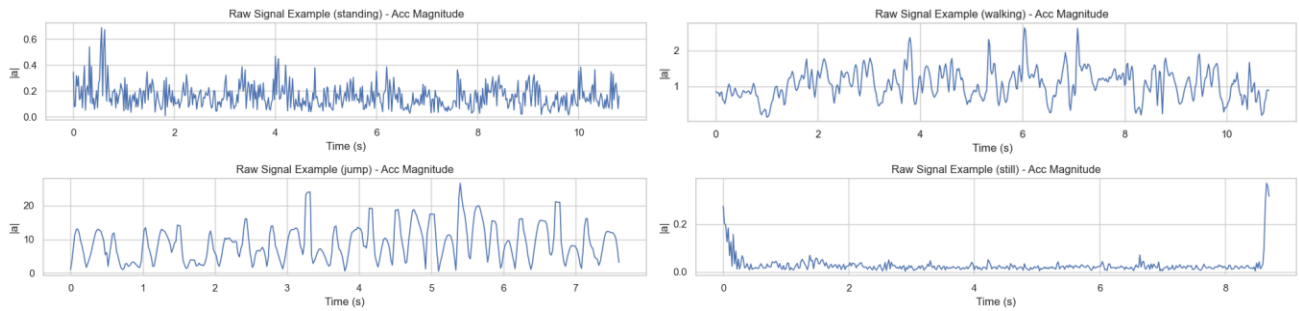

Figure 1: Accelerometer magnitude for all four activities (harmonized to 50 Hz). Jumping shows high-amplitude impulses; still is near-zero; walking is periodic; standing shows low-level postural sway.

2.2 Windowing and Feature Extraction

Each recording was segmented into **2-second windows (100 samples) with 50% overlap**. The 2 s duration captures at least two full walking strides, while the 50% overlap reduces boundary artefacts and increases training sample count. Twenty features were extracted per window spanning both domains:

Time-domain: mean and variance per accelerometer axis, signal magnitude area (SMA), RMS of |a|, mean and variance of gyroscope magnitude |g|.

Frequency-domain (FFT): dominant frequency of |a| and |g|, spectral energy of |a| and |g|, cross-axis correlations (acc_xy, acc_xz, acc_yz). All 20 features were **Z-score normalized** using statistics fit on the training set only.

3. HMM Setup and Implementation

3.1 Model Structure

Component	Symbol	Description
Hidden States	Z	4 activities: standing, walking, jumping, still
Observations	X	20-dim feature vectors (time + frequency domain)
Transition Matrix	A (4×4)	Probability of moving between activities across successive windows
Emission	B (Gaussian)	Diagonal-covariance Gaussian per state; means and variances learned via Baum-Welch
Initial Probs.	π (4×1)	Probability of each activity being the first state; initialized from training counts

3.2 Baum-Welch Training and Convergence

The Baum-Welch EM algorithm was implemented from scratch in NumPy using **log-domain arithmetic** throughout (custom logsumexp). Parameters were initialized from supervised training statistics. Training ran for up to 80 iterations with early stopping when Δlog-likelihood < 10^{-3}. As shown in Figure 2, convergence was achieved in **~7 iterations**, with log-likelihood rising from ~2000 to ~2455 and plateauing — confirming correct EM implementation.

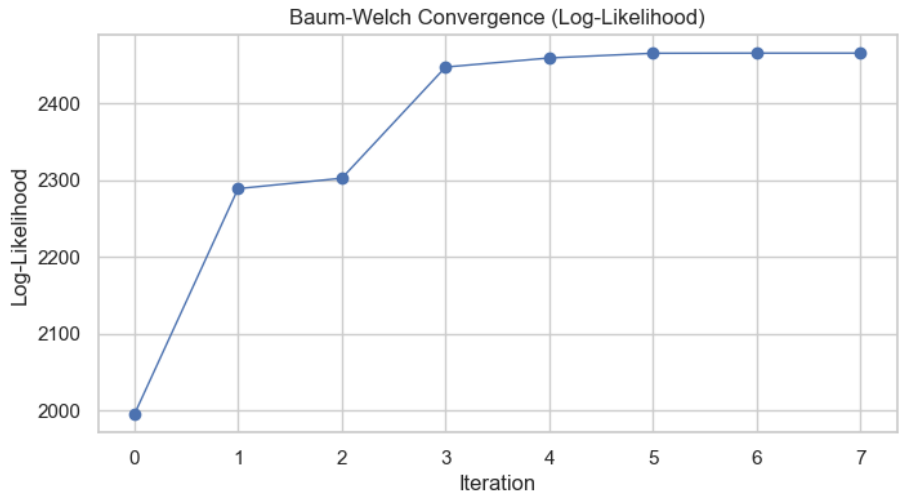

Figure 2: Baum-Welch training convergence. Log-likelihood increases monotonically and plateaus after ~7 iterations.

3.3 Viterbi Decoding

The Viterbi algorithm, also implemented in log space from scratch, decodes the most probable hidden state sequence for a given observation sequence. Since HMM state indices are arbitrary, a greedy one-to-one mapping from hidden states to activity labels was derived on the training set and applied consistently at test time.

4. Results and Interpretation

4.1 Transition Matrix and Confusion Matrix

Figure 3 (left) shows the learned transition matrix. The near-perfect diagonal (0.97–1.00) confirms that the model has learned strong activity persistence — once an activity begins, it continues across successive 2-second windows. The small off-diagonal values (e.g., walking → jump = 0.01) reflect realistic but rare transitions. Figure 3 (right) shows the confusion matrix on the unseen test set: all 61 windows are correctly classified, yielding 100% accuracy with zero off-diagonal errors.

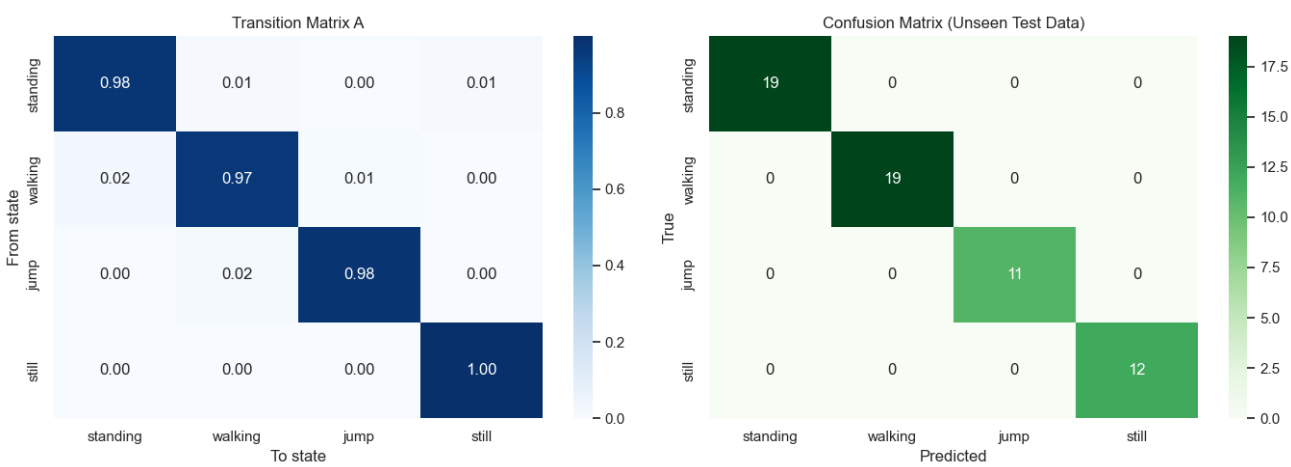

Figure 3: (Left) Learned transition matrix — strong diagonal dominance reflects activity persistence. (Right) Confusion matrix on unseen test data — perfect classification across all four activities.

4.2 Emission Parameters

Figure 4 shows the Gaussian emission means for the top 10 discriminative features. State 3 (high-intensity activity) exhibits strongly elevated acc_sma, gyr_std, acc_spec_energy, acc_std, and gyr_var — consistent with jumping.

State 0 shows a distinctive positive corr_acc_xy value. States 1 and 2 occupy intermediate feature ranges corresponding to walking and standing respectively.

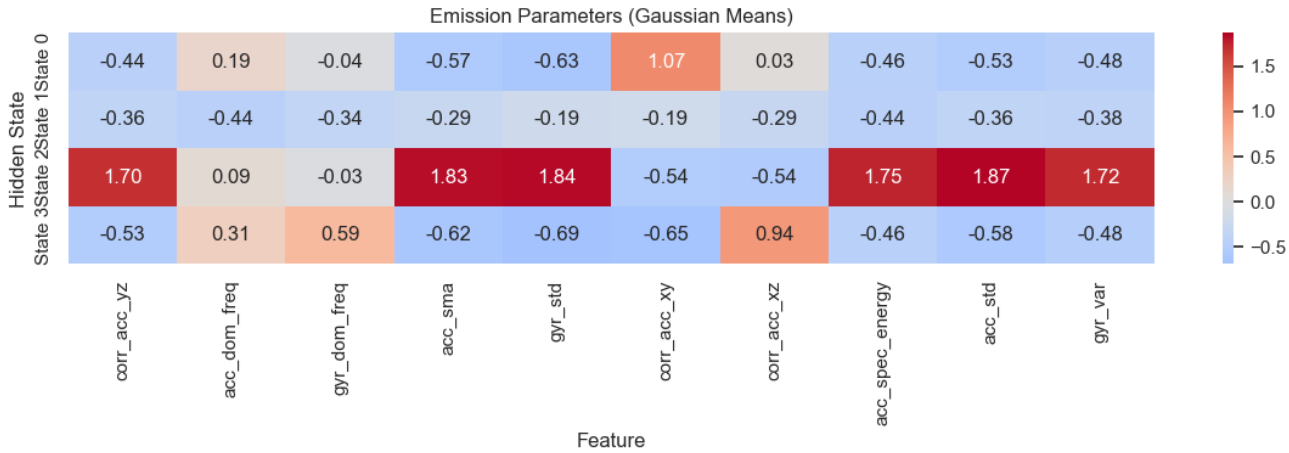

Figure 4: Emission parameter heatmap (Gaussian means, top 10 features by training variance). Red = high, blue = low. State 3 (jumping) dominates high-energy features.

4.3 Decoded Activity Sequence

Figure 5 shows the Viterbi-decoded sequence for an unseen jumping test file. The true and predicted sequences are perfectly overlapping, demonstrating that the HMM correctly identifies the jumping activity across all windows with no misclassification.

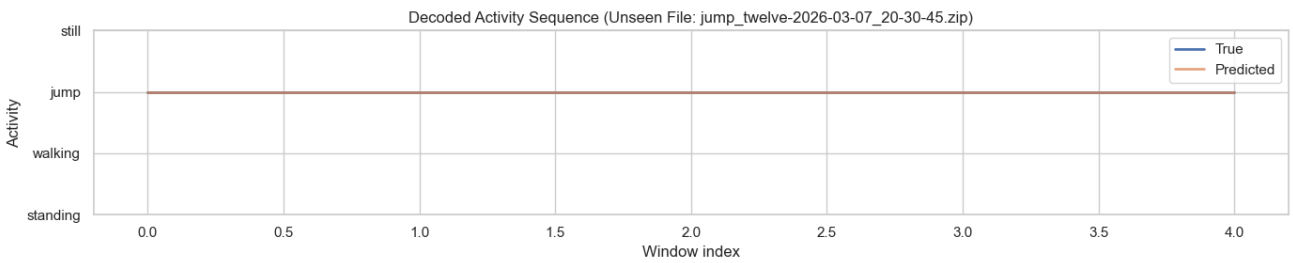

Figure 5: Decoded activity sequence for unseen test file (jump). True labels (blue) and Viterbi predictions (orange) overlap perfectly.

4.4 Quantitative Metrics

Activity	# Samples	Sensitivity	Specificity	Overall Accuracy
Standing	19	1.00	1.00	1.00
Walking	19	1.00	1.00	1.00
Jumping	11	1.00	1.00	1.00
Still	12	1.00	1.00	1.00

Table 1: Per-activity evaluation metrics on unseen test data (61 total windows). Overall accuracy = 1.00.

5. Discussion and Conclusion

The HMM achieved **100% accuracy** on the unseen test set, with perfect sensitivity and specificity per class. The most physically interpretable result is the strong diagonal of the transition matrix, which reflects that activities persist for multiple 2-second windows — a realistic pattern in human movement. The emission heatmap confirms that the model has learned meaningful per-activity feature signatures: jumping dominates energy-based features, still is near-zero across all features, and walking occupies a moderate-energy, low-frequency band.

The combination of frequency-domain features (dominant frequency, spectral energy) with time-domain features (SMA, RMS, variance) provided sufficient discriminability for clean separation of all four activities, even across two different devices. Z-score normalization on training data only ensured no data leakage to the test set.

Potential improvements include: (1) collecting data from more participants for better generalization; (2) adding spectral entropy or wavelet features to improve standing/walking distinction at low speeds; (3) using full-covariance or Gaussian mixture emissions for richer state modelling; (4) incorporating barometric pressure or magnetometer signals as additional discriminative sources.

6. Collaboration and Task Allocation

Member	Activities	Technical Work	Evidence
nshimiyeemmy	Jumping, Still	Data collection, notebook baseline, feature pipeline, HMM training + evaluation	4 dataset commits + baseline notebook commit
hirwaalu	Standing, Walking	Data collection, report writing, task allocation, submission structure	4 dataset commits + notebook enhancement commit